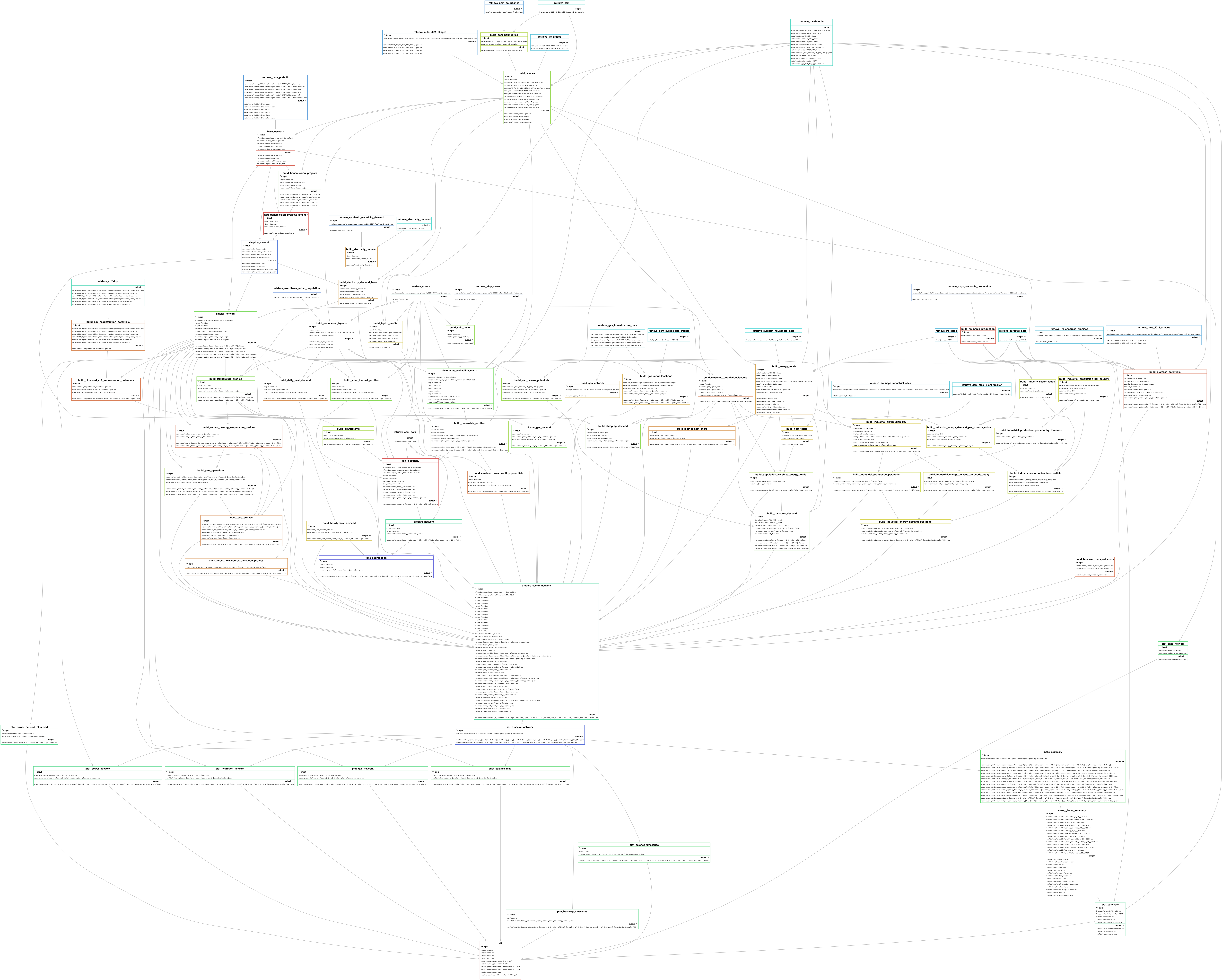
retrieve_osm_boundaries
output →
data/osm-boundaries/json/{country}_adm1.json
retrieve_eez
output →
data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg
retrieve_nuts_2021_shapes
↪ input
.snakemake/storage/http/gisco-services.ec.europa.eu/distribution/v2/nuts/download/ref-nuts-2021-01m.geojson.zip
output →
data/nuts/NUTS_RG_01M_2021_4326_LEVL_0.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_1.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_2.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_3.geojson
build_osm_boundaries
↪ input
data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg
data/osm-boundaries/json/{country}_adm1.json
output →
data/osm-boundaries/build/{country}_adm1.geojson
retrieve_jrc_ardeco
output →
data/jrc-ardeco/ARDECO-SNPTD.2021.table.csv
data/jrc-ardeco/ARDECO-SUVGDP.2021.table.csv
retrieve_databundle
output →
data/bundle/GDP_per_capita_PPP_1990_2015_v2.nc
data/bundle/corine/g250_clc06_V18_5.tif
data/bundle/eea/UNFCCC_v23.csv
data/bundle/emobility/KFZ__count
data/bundle/emobility/Pkw__count
data/bundle/era5-HDD-per-country.csv
data/bundle/era5-runoff-per-country.csv
data/bundle/gebco/GEBCO_2014_2D.nc
data/bundle/h2_salt_caverns_GWh_per_sqkm.geojson
data/bundle/je-e-21.03.02.xls
data/bundle/nama_10r_3popgdp.tsv.gz
data/bundle/natura/natura.tiff
data/bundle/ppp_2019_1km_Aggregated.tif
build_shapes
↪ input
<input function>
data/bundle/GDP_per_capita_PPP_1990_2015_v2.nc
data/bundle/ppp_2019_1km_Aggregated.tif
data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg
data/jrc-ardeco/ARDECO-SNPTD.2021.table.csv
data/jrc-ardeco/ARDECO-SUVGDP.2021.table.csv
data/nuts/NUTS_RG_01M_2021_4326_LEVL_3.geojson
data/osm-boundaries/build/BA_adm1.geojson
data/osm-boundaries/build/MD_adm1.geojson
data/osm-boundaries/build/UA_adm1.geojson
data/osm-boundaries/build/XK_adm1.geojson
output →
resources/country_shapes.geojson
resources/europe_shape.geojson
resources/nuts3_shapes.geojson
resources/offshore_shapes.geojson
retrieve_osm_prebuilt
↪ input
.snakemake/storage/http/zenodo.org/records/14144752/files/buses.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/converters.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/lines.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/links.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/map.html
.snakemake/storage/http/zenodo.org/records/14144752/files/transformers.csv
output →
data/osm-prebuilt/0.6/buses.csv
data/osm-prebuilt/0.6/converters.csv
data/osm-prebuilt/0.6/lines.csv
data/osm-prebuilt/0.6/links.csv
data/osm-prebuilt/0.6/map.html
data/osm-prebuilt/0.6/transformers.csv
base_network
↪ input
<function input_base_network at 0x14ecfea20>
resources/country_shapes.geojson
resources/europe_shape.geojson
resources/nuts3_shapes.geojson
resources/offshore_shapes.geojson
output →
resources/admin_shapes.geojson
resources/networks/base.nc
resources/regions_offshore.geojson
resources/regions_onshore.geojson
build_transmission_projects
↪ input
<input function>
resources/europe_shape.geojson
resources/networks/base.nc
resources/offshore_shapes.geojson
output →
resources/transmission_projects/adjust_lines.csv
resources/transmission_projects/adjust_links.csv
resources/transmission_projects/new_buses.csv
resources/transmission_projects/new_lines.csv
resources/transmission_projects/new_links.csv
add_transmission_projects_and_dlr
↪ input
<input function>
<input function>
resources/networks/base.nc
output →
resources/networks/base_extended.nc
retrieve_synthetic_electricity_demand
↪ input
.snakemake/storage/http/zenodo.org/records/10820928/files/demand_hourly.csv
output →
data/load_synthetic_raw.csv
retrieve_electricity_demand
output →
data/electricity_demand_raw.csv
simplify_network
↪ input
resources/admin_shapes.geojson
resources/networks/base_extended.nc
resources/regions_offshore.geojson
resources/regions_onshore.geojson
output →
resources/busmap_base_s.csv
resources/networks/base_s.nc
resources/regions_offshore_base_s.geojson
resources/regions_onshore_base_s.geojson
build_electricity_demand
↪ input
<input function>
data/electricity_demand_raw.csv
output →
resources/electricity_demand.csv
retrieve_co2stop
output →
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Storage_Units.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps1.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps_Temp.csv
data/CO2JRC_OpenFormats/CO2Stop_Polygons Data/DaughterUnits_March13.kml
data/CO2JRC_OpenFormats/CO2Stop_Polygons Data/StorageUnits_March13.kml
retrieve_worldbank_urban_population
output →
data/worldbank/API_SP.URB.TOTL.IN.ZS_DS2_en_csv_v2.csv
build_electricity_demand_base
↪ input
resources/electricity_demand.csv
resources/networks/base_s.nc
resources/nuts3_shapes.geojson
resources/regions_onshore_base_s.geojson
output →
resources/electricity_demand_base_s.nc
retrieve_cutout
↪ input
.snakemake/storage/http/zenodo.org/records/15349674/files/{cutout}.nc
output →
cutouts/{cutout}.nc
retrieve_ship_raster
↪ input
.snakemake/storage/http/zenodo.org/records/13757228/files/shipdensity_global.zip
output →
data/shipdensity_global.zip
build_co2_sequestration_potentials
↪ input
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Storage_Units.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps1.csv
data/CO2JRC_OpenFormats/CO2Stop_DataInterrogationSystem/Hydrocarbon_Traps_Temp.csv
data/CO2JRC_OpenFormats/CO2Stop_Polygons Data/DaughterUnits_March13.kml
data/CO2JRC_OpenFormats/CO2Stop_Polygons Data/StorageUnits_March13.kml
output →
resources/co2_sequestration_potentials.geojson
cluster_network
↪ input
<function input_custom_busmap at 0x14ed5b060>
<input function>
<input function>
resources/admin_shapes.geojson
resources/electricity_demand_base_s.nc
resources/networks/base_s.nc
resources/regions_offshore_base_s.geojson
resources/regions_onshore_base_s.geojson
output →
resources/busmap_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/linemap_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
resources/regions_offshore_base_s_{clusters,[0-9]+(m|c)?|all|adm}.geojson
resources/regions_onshore_base_s_{clusters,[0-9]+(m|c)?|all|adm}.geojson
build_population_layouts
↪ input
<input function>
data/worldbank/API_SP.URB.TOTL.IN.ZS_DS2_en_csv_v2.csv
resources/nuts3_shapes.geojson
output →
resources/pop_layout_rural.nc
resources/pop_layout_total.nc
resources/pop_layout_urban.nc
build_hydro_profile
↪ input
<input function>
data/bundle/era5-runoff-per-country.csv
data/eia_hydro_annual_capacity.csv
data/eia_hydro_annual_generation.csv
resources/country_shapes.geojson
output →
resources/profile_hydro.nc
build_ship_raster
↪ input
<input function>
data/shipdensity_global.zip
output →
resources/shipdensity_raster.tif
retrieve_gas_infrastructure_data
output →
data/gas_network/scigrid-gas/data/IGGIELGN_BorderPoints.geojson
data/gas_network/scigrid-gas/data/IGGIELGN_LNGs.geojson
data/gas_network/scigrid-gas/data/IGGIELGN_PipeSegments.geojson
data/gas_network/scigrid-gas/data/IGGIELGN_Productions.geojson
data/gas_network/scigrid-gas/data/IGGIELGN_Storages.geojson
retrieve_gem_europe_gas_tracker
output →
data/gem/Europe-Gas-Tracker-2024-05.xlsx
retrieve_eurostat_household_data
output →
data/eurostat/eurostat-household_energy_balances-february_2024.csv
retrieve_usgs_ammonia_production
↪ input
.snakemake/storage/http/d9-wret.s3.us-west-2.amazonaws.com/assets/palladium/production/s3fs-public/media/files/myb1-2022-nitro-ert.xlsx
output →
data/myb1-2022-nitro-ert.xlsx
retrieve_jrc_idees
output →
data/jrc-idees-2021
build_ammonia_production
↪ input
data/myb1-2022-nitro-ert.xlsx
output →
resources/ammonia_production.csv
retrieve_eurostat_data
output →
data/eurostat/Balances-April2023
retrieve_jrc_enspreso_biomass
↪ input
.snakemake/storage/http/zenodo.org/records/10356004/files/ENSPRESO_BIOMASS.xlsx
output →
data/ENSPRESO_BIOMASS.xlsx
retrieve_nuts_2013_shapes
↪ input
.snakemake/storage/http/gisco-services.ec.europa.eu/distribution/v2/nuts/download/ref-nuts-2013-03m.geojson.zip
output →
data/nuts/NUTS_RG_03M_2013_4326_LEVL_2.geojson
data/nuts/NUTS_RG_03M_2013_4326_LEVL_3.geojson
build_clustered_co2_sequestration_potentials
↪ input
resources/co2_sequestration_potentials.geojson
resources/regions_offshore_base_s_{clusters}.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/co2_sequestration_potential_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_temperature_profiles
↪ input
<input function>
resources/pop_layout_total.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/temp_air_total_base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
resources/temp_soil_total_base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
build_daily_heat_demand
↪ input
<input function>
resources/pop_layout_total.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/daily_heat_demand_total_base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
build_solar_thermal_profiles
↪ input
<input function>
resources/pop_layout_total.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/solar_thermal_total_base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
determine_availability_matrix
↪ input
<function <lambda> at 0x14ed5b4c0>
<function input_ua_md_availability_matrix at 0x14ed5a3e0>
<input function>
<input function>
<input function>
<input function>
<input function>
data/bundle/corine/g250_clc06_V18_5.tif
resources/country_shapes.geojson
resources/offshore_shapes.geojson
output →
resources/availability_matrix_{clusters,[0-9]+(m|c)?|all|adm}_{technology}.nc
build_salt_cavern_potentials
↪ input
data/bundle/h2_salt_caverns_GWh_per_sqkm.geojson
resources/regions_offshore_base_s_{clusters}.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/salt_cavern_potentials_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_gas_network
↪ input
data/gas_network/scigrid-gas/data/IGGIELGN_PipeSegments.geojson
output →
resources/gas_network.csv
build_gas_input_locations
↪ input
data/gas_network/scigrid-gas/data/IGGIELGN_BorderPoints.geojson
data/gas_network/scigrid-gas/data/IGGIELGN_Storages.geojson
data/gem/Europe-Gas-Tracker-2024-05.xlsx
resources/regions_offshore_base_s_{clusters}.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/gas_input_locations_s_{clusters,[0-9]+(m|c)?|all|adm}.geojson
resources/gas_input_locations_s_{clusters,[0-9]+(m|c)?|all|adm}_simplified.csv
build_clustered_population_layouts
↪ input
<input function>
resources/pop_layout_rural.nc
resources/pop_layout_total.nc
resources/pop_layout_urban.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/pop_layout_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_energy_totals
↪ input
data/bundle/eea/UNFCCC_v23.csv
data/district_heat_share.csv
data/eurostat/Balances-April2023
data/eurostat/eurostat-household_energy_balances-february_2024.csv
data/gr-e-11.03.02.01.01-cc.csv
data/jrc-idees-2021
data/switzerland-new_format-all_years.csv
resources/nuts3_shapes.geojson
output →
resources/co2_totals.csv
resources/district_heat_share.csv
resources/energy_totals.csv
resources/heating_efficiencies.csv
resources/transformation_output_coke.csv
resources/transport_data.csv
retrieve_hotmaps_industrial_sites
↪ input
.snakemake/storage/http/gitlab.com/hotmaps/industrial_sites/industrial_sites_Industrial_Database/-/raw/master/data/Industrial_Database.csv
output →
data/Industrial_Database.csv
retrieve_gem_steel_plant_tracker
output →
data/gem/Global-Steel-Plant-Tracker-April-2024-Standard-Copy-V1.xlsx
build_industry_sector_ratios
↪ input
data/jrc-idees-2021
resources/ammonia_production.csv
output →
resources/industry_sector_ratios.csv
build_industrial_production_per_country
↪ input
data/ch_industrial_production_per_subsector.csv
data/eurostat/Balances-April2023
data/jrc-idees-2021
resources/ammonia_production.csv
output →
resources/industrial_production_per_country.csv
build_biomass_potentials
↪ input
data/ENSPRESO_BIOMASS.xlsx
data/bundle/je-e-21.03.02.xls
data/bundle/nama_10r_3popgdp.tsv.gz
data/ch_cantons.csv
data/eurostat/Balances-April2023
data/nuts/NUTS_RG_03M_2013_4326_LEVL_2.geojson
resources/country_shapes.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/biomass_potentials_all_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.csv
resources/biomass_potentials_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.csv
build_central_heating_temperature_profiles
↪ input
resources/regions_onshore_base_s_{clusters}.geojson
resources/temp_air_total_base_s_{clusters}.nc
output →
resources/central_heating_forward_temperature_profiles_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
resources/central_heating_return_temperature_profiles_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
build_powerplants
↪ input
data/custom_powerplants.csv
resources/networks/base_s_{clusters}.nc
output →
resources/powerplants_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
retrieve_cost_data
output →
resources/costs_{year}.csv
build_renewable_profiles
↪ input
<input function>
<input function>
resources/availability_matrix_{clusters}_{technology}.nc
resources/offshore_shapes.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/profile_{clusters,[0-9]+(m|c)?|all|adm}_{technology,(?!hydro).*}.nc
resources/regions_by_class_{clusters,[0-9]+(m|c)?|all|adm}_{technology,(?!hydro).*}.geojson
cluster_gas_network
↪ input
resources/gas_network.csv
resources/regions_offshore_base_s_{clusters}.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/gas_network_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_shipping_demand
↪ input
data/attributed_ports.json
resources/energy_totals.csv
resources/europe_shape.geojson
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/shipping_demand_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_district_heat_share
↪ input
resources/district_heat_share.csv
resources/pop_layout_base_s_{clusters}.csv
output →
resources/district_heat_share_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.csv
build_heat_totals
↪ input
data/bundle/era5-HDD-per-country.csv
resources/energy_totals.csv
output →
resources/heat_totals.csv
build_industrial_distribution_key
↪ input
data/Industrial_Database.csv
data/ammonia_plants.csv
data/cement-plants-noneu.csv
data/gem/Global-Steel-Plant-Tracker-April-2024-Standard-Copy-V1.xlsx
data/refineries-noneu.csv
resources/pop_layout_base_s_{clusters}.csv
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/industrial_distribution_key_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_industrial_energy_demand_per_country_today
↪ input
data/jrc-idees-2021
resources/industrial_production_per_country.csv
resources/transformation_output_coke.csv
output →
resources/industrial_energy_demand_per_country_today.csv
build_industrial_production_per_country_tomorrow
↪ input
resources/industrial_production_per_country.csv
output →
resources/industrial_production_per_country_tomorrow_{planning_horizons,[0-9]{4}}.csv
build_ptes_operations
↪ input
resources/central_heating_forward_temperature_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/central_heating_return_temperature_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/ptes_direct_utilisation_profiles_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
resources/ptes_e_max_pu_profiles_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
resources/ptes_top_temperature_profiles_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
add_electricity
↪ input
<function input_class_regions at 0x14ed5a660>
<function input_conventional at 0x14ed5bec0>
<function input_profile_tech at 0x14ed5bc40>
<input function>
<input function>
data/hydro_capacities.csv
data/unit_commitment.csv
resources/busmap_base_s_{clusters}.csv
resources/electricity_demand_base_s.nc
resources/networks/base_s_{clusters}.nc
resources/powerplants_s_{clusters}.csv
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_elec.nc
build_clustered_solar_rooftop_potentials
↪ input
<input function>
resources/pop_layout_total.nc
resources/regions_by_class_{clusters}_solar.geojson
output →
resources/solar_rooftop_potentials_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_population_weighted_energy_totals
↪ input
resources/pop_layout_base_s_{clusters}.csv
resources/{kind}_totals.csv
output →
resources/pop_weighted_{kind}_totals_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_industrial_production_per_node
↪ input
resources/industrial_distribution_key_base_s_{clusters}.csv
resources/industrial_production_per_country_tomorrow_{planning_horizons}.csv
output →
resources/industrial_production_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.csv
build_industrial_energy_demand_per_node_today
↪ input
resources/industrial_distribution_key_base_s_{clusters}.csv
resources/industrial_energy_demand_per_country_today.csv
output →
resources/industrial_energy_demand_today_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_industry_sector_ratios_intermediate
↪ input
resources/industrial_energy_demand_per_country_today.csv
resources/industrial_production_per_country.csv
resources/industry_sector_ratios.csv
output →
resources/industry_sector_ratios_{planning_horizons,[0-9]{4}}.csv
build_cop_profiles
↪ input
resources/central_heating_forward_temperature_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/central_heating_return_temperature_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/ptes_top_temperature_profiles_s_{clusters}_{planning_horizons}.nc
resources/regions_onshore_base_s_{clusters}.geojson
resources/temp_air_total_base_s_{clusters}.nc
resources/temp_soil_total_base_s_{clusters}.nc
output →
resources/cop_profiles_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
build_hourly_heat_demand
↪ input
data/heat_load_profile_BDEW.csv
resources/daily_heat_demand_total_base_s_{clusters}.nc
output →
resources/hourly_heat_demand_total_base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
prepare_network
↪ input
<input function>
<input function>
resources/networks/base_s_{clusters}_elec.nc
output →
resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_elec_{opts,[-+a-zA-Z0-9\.]*}.nc
build_transport_demand
↪ input
data/bundle/emobility/KFZ__count
data/bundle/emobility/Pkw__count
resources/pop_layout_base_s_{clusters}.csv
resources/pop_weighted_energy_totals_s_{clusters}.csv
resources/temp_air_total_base_s_{clusters}.nc
resources/transport_data.csv
output →
resources/avail_profile_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/dsm_profile_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/transport_data_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/transport_demand_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
build_industrial_energy_demand_per_node
↪ input
resources/industrial_energy_demand_today_base_s_{clusters}.csv
resources/industrial_production_base_s_{clusters}_{planning_horizons}.csv
resources/industry_sector_ratios_{planning_horizons}.csv
output →
resources/industrial_energy_demand_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.csv
build_direct_heat_source_utilisation_profiles
↪ input
resources/central_heating_forward_temperature_profiles_base_s_{clusters}_{planning_horizons}.nc
output →
resources/direct_heat_source_utilisation_profiles_base_s_{clusters,[0-9]+(m|c)?|all|adm}_{planning_horizons,[0-9]{4}}.nc
time_aggregation
↪ input
<input function>
<input function>
resources/networks/base_s_{clusters}_elec_{opts}.nc
output →
resources/snapshot_weightings_base_s_{clusters,[0-9]+(m|c)?|all|adm}_elec_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}.csv
build_biomass_transport_costs
↪ input
data/biomass_transport_costs_supplychain1.csv
data/biomass_transport_costs_supplychain2.csv
output →
resources/biomass_transport_costs.csv
prepare_sector_network
↪ input
<function input_heat_source_power at 0x14eed9800>
<function input_profile_offwind at 0x14eed89a0>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
<input function>
data/bundle/eea/UNFCCC_v23.csv
data/eurostat/Balances-April2023
resources/avail_profile_s_{clusters}.csv
resources/biomass_potentials_s_{clusters}_{planning_horizons}.csv
resources/busmap_base_s.csv
resources/busmap_base_s_{clusters}.csv
resources/co2_totals.csv
resources/cop_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/direct_heat_source_utilisation_profiles_base_s_{clusters}_{planning_horizons}.nc
resources/district_heat_share_base_s_{clusters}_{planning_horizons}.csv
resources/dsm_profile_s_{clusters}.csv
resources/gas_input_locations_s_{clusters}.geojson
resources/gas_input_locations_s_{clusters}_simplified.csv
resources/gas_network_base_s_{clusters}.csv
resources/heating_efficiencies.csv
resources/hourly_heat_demand_total_base_s_{clusters}.nc
resources/industrial_energy_demand_base_s_{clusters}_{planning_horizons}.csv
resources/industrial_production_base_s_{clusters}_{planning_horizons}.csv
resources/networks/base_s_{clusters}_elec_{opts}.nc
resources/pop_layout_base_s_{clusters}.csv
resources/pop_weighted_energy_totals_s_{clusters}.csv
resources/pop_weighted_heat_totals_s_{clusters}.csv
resources/salt_cavern_potentials_s_{clusters}.csv
resources/shipping_demand_s_{clusters}.csv
resources/snapshot_weightings_base_s_{clusters}_elec_{opts}_{sector_opts}.csv
resources/temp_air_total_base_s_{clusters}.nc
resources/temp_soil_total_base_s_{clusters}.nc
resources/transport_data_s_{clusters}.csv
resources/transport_demand_s_{clusters}.csv
output →
resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.nc
plot_base_network
↪ input
resources/networks/base.nc
resources/regions_onshore.geojson
output →
resources/maps/power-network.pdf
plot_power_network_clustered
↪ input
resources/networks/base_s_{clusters}.nc
resources/regions_onshore_base_s_{clusters}.geojson
output →
resources/maps/power-network-s-{clusters,[0-9]+(m|c)?|all|adm}.pdf
solve_sector_network
↪ input
resources/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/configs/config.base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.yaml
results/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.nc
make_summary
↪ input
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/csvs/individual/capacities_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/capacity_factors_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/costs_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/curtailment_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/energy_balance_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/energy_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/market_values_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/metrics_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/nodal_capacities_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/nodal_capacity_factors_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/nodal_costs_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/nodal_energy_balance_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/prices_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
results/csvs/individual/weighted_prices_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}.csv
plot_power_network
↪ input
resources/regions_onshore_base_s_{clusters}.geojson
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/maps/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}-costs-all_{planning_horizons,[0-9]{4}}.pdf
plot_hydrogen_network
↪ input
resources/regions_onshore_base_s_{clusters}.geojson
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/maps/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}-h2_network_{planning_horizons,[0-9]{4}}.pdf
plot_gas_network
↪ input
resources/regions_onshore_base_s_{clusters}.geojson
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/maps/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}-ch4_network_{planning_horizons,[0-9]{4}}.pdf
plot_balance_map
↪ input
resources/regions_onshore_base_s_{clusters}.geojson
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/maps/base_s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}-balance_map_{carrier}.pdf
make_global_summary
↪ input
results/csvs/individual/capacities_s_50___2050.csv
results/csvs/individual/capacity_factors_s_50___2050.csv
results/csvs/individual/costs_s_50___2050.csv
results/csvs/individual/curtailment_s_50___2050.csv
results/csvs/individual/energy_balance_s_50___2050.csv
results/csvs/individual/energy_s_50___2050.csv
results/csvs/individual/market_values_s_50___2050.csv
results/csvs/individual/metrics_s_50___2050.csv
results/csvs/individual/nodal_capacities_s_50___2050.csv
results/csvs/individual/nodal_capacity_factors_s_50___2050.csv
results/csvs/individual/nodal_costs_s_50___2050.csv
results/csvs/individual/nodal_energy_balance_s_50___2050.csv
results/csvs/individual/prices_s_50___2050.csv
results/csvs/individual/weighted_prices_s_50___2050.csv
output →
results/csvs/capacities.csv
results/csvs/capacity_factors.csv
results/csvs/costs.csv
results/csvs/curtailment.csv
results/csvs/energy.csv
results/csvs/energy_balance.csv
results/csvs/market_values.csv
results/csvs/metrics.csv
results/csvs/nodal_capacities.csv
results/csvs/nodal_capacity_factors.csv
results/csvs/nodal_costs.csv
results/csvs/nodal_energy_balance.csv
results/csvs/prices.csv
results/csvs/weighted_prices.csv
plot_balance_timeseries
↪ input
matplotlibrc
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/graphics/balance_timeseries/s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}
plot_heatmap_timeseries
↪ input
matplotlibrc
results/networks/base_s_{clusters}_{opts}_{sector_opts}_{planning_horizons}.nc
output →
results/graphics/heatmap_timeseries/s_{clusters,[0-9]+(m|c)?|all|adm}_{opts,[-+a-zA-Z0-9\.]*}_{sector_opts,[-+a-zA-Z0-9\.\s]*}_{planning_horizons,[0-9]{4}}
plot_summary
↪ input
data/bundle/eea/UNFCCC_v23.csv
data/eurostat/Balances-April2023
results/csvs/costs.csv
results/csvs/energy.csv
results/csvs/energy_balance.csv
output →
results/graphs/balances-energy.svg
results/graphs/costs.svg
results/graphs/energy.svg
all
↪ input
<input function>
<input function>
<input function>
<input function>
resources/maps/power-network-s-50.pdf
resources/maps/power-network.pdf
results/graphics/balance_timeseries/s_50___2050
results/graphics/heatmap_timeseries/s_50___2050
results/graphs/costs.svg
results/maps/base_s_50__-costs-all_2050.pdf